

SAI KUKKAPALLI

Software Engineer – DevOps | Cloud Native Infrastructure | CI/CD Automation

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PROFESSIONAL SUMMARY

B.Tech Computer Science graduate (2026) and aspiring Software Engineer - DevOps with hands-on experience building CI/CD pipelines, containerizing applications with Docker, and automating AWS cloud infrastructure using Infrastructure-as-Code practices. Comfortable writing Python code, designing REST APIs with MVC principles, and building automated lint/test gates for testability. Reduced Docker image size by 40% using multi-stage builds and automated 100% of deployment workflows across project pipelines. Actively deepening expertise in Kubernetes, Terraform, Ansible, and observability tooling (Prometheus, Grafana). Collaborative and detail-oriented, with a strong interest in large-scale automation and AI-assisted development tooling - eager to learn from senior engineers and contribute to cloud-native, distributed systems.

TECHNICAL SKILLS

Cloud: AWS (EC2, S3, IAM, CloudFront, VPC, Route 53)

Containers: Docker, Docker Compose, Multi-stage Dockerfile

Orchestration: Kubernetes, Docker Swarm

Monitoring: Prometheus, Grafana (fundamentals)

Version Control: Git, GitHub, Branching Strategies, PRs

APIs & Architecture: REST APIs, MVC, Microservices (fundamentals)

CI/CD: GitHub Actions, Jenkins

IaC & Config: Terraform, CloudFormation, Ansible

Scripting: Python, Bash/Shell, YAML

Networking: DNS, Load Balancing, CORS, CDN, HTTPS

OS & Practices: Linux (Ubuntu, CentOS), DevSecOps, Agile/Scrum

Testing & Quality: Unit Testing, Linting, PR Code Reviews

DEVOPS PROJECTS

Flask To-Do App — Containerized CI/CD Pipeline | *Personal Project*

Jan 2025 – Mar 2025

Tech Stack: Python Flask, Docker, Docker Compose, GitHub Actions, REST API, Git

- Architected a full-stack RESTful To-Do application using Python Flask with PostgreSQL integration, MVC design pattern, and a clean REST API surface.
- Built an end-to-end CI/CD pipeline with GitHub Actions — automated linting, unit tests, Docker image builds, and deployments on every pull request, supporting testability and achieving zero-downtime releases.
- Containerized the app using Docker multi-stage builds, cutting the final image size by 40% and eliminating environment inconsistencies across dev/staging/prod — an Infrastructure-as-Code approach to environment parity.
- Implemented a structured Git branching strategy (feature → dev → main) with enforced PR reviews and automated linting gates, improving code quality and team-readiness.
- Configured Docker Compose for local multi-service orchestration (app + DB + reverse proxy) in a microservices-style architecture, enabling consistent one-command developer environments.

Automated Static Website Deployment — AWS S3 + CloudFront |

Nov 2024 – Jan 2025

Personal Project

Tech Stack: AWS S3, CloudFront, IAM, Route 53, GitHub Actions, YAML

- Hosted a secure, production-ready static website on AWS S3 with global availability, leveraging HTTPS for encrypted access and reliable performance.
- Automated the full deployment pipeline with GitHub Actions — commits to main trigger build, test, and S3 sync within 2 minutes, enabling continuous delivery.
- Applied IAM Least Privilege principles as an early DevSecOps practice: scoped IAM roles for the CI pipeline, eliminating long-lived credentials and reducing the blast radius of a potential key compromise.

- Configured S3 bucket policies, CORS rules, static website hosting, and CloudFront cache invalidation for seamless content updates.
- Eliminated 100% of manual deployment effort — replaced 15-minute manual FTP uploads with a fully automated push-to-deploy workflow.

EDUCATION

B.Tech — Computer Science & Engineering | *PBR Visvodaya Institute of Technology & Science, Nellore, AP*

2022 – 2026

CGPA: 7.06 /10 | **Relevant Coursework:** Operating Systems, Computer Networks, Cloud Computing, DBMS, Data Structures & Algorithms, Software Engineering

CERTIFICATIONS & TRAINING

- Git & GitHub Complete Course — Version control, branching strategies, CI/CD integration, collaborative workflows (2024)
- AWS Cloud Practitioner Essentials — EC2, S3, IAM, CloudFront, VPC, cloud infrastructure fundamentals (Self-study, 2024)
- Docker & Containerization — Container lifecycle, Dockerfile optimization, Docker Compose, image registries (2024)

KEY ACHIEVEMENTS

- Reduced Docker image size by 40% using multi-stage builds, improving deployment speed and reducing cloud storage costs across projects.
- Automated 100% of deployment workflows with GitHub Actions — eliminated all manual release steps across 2 production-grade projects.
- Dedicated 2–3 hours daily to upskilling on Kubernetes, Terraform, Ansible, AWS advanced services, observability tooling, and SRE practices.
- Solid Linux fundamentals: file system management, process control, networking, shell scripting, cron automation, and system hardening.
- Agile/Scrum practitioner with experience in sprint planning, code reviews, and PR-based collaborative development — receptive to feedback and mentorship from senior engineers.

Open to DevOps/Software Engineering Internships & Entry-Level Roles | Available Immediately | Open to Onsite Work & Relocation